

# FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to [grants@foundationfar.org](mailto:grants@foundationfar.org).

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

| Grant Information                                   |                                                        |
|-----------------------------------------------------|--------------------------------------------------------|
| Grant ID                                            | DSnew-0000000017                                       |
| Award Program                                       | Direct Solicitation without Pre-Proposal               |
| Project Title                                       | Food and Agricultural Vulnerability Index              |
| Grant Start Date                                    | 08/01/2020                                             |
| Grant End Date                                      | 07/31/2022                                             |
| Total Grant Budget                                  | \$221,743.33                                           |
| Total FFAR Budget                                   | \$221,743.33                                           |
| Matching Funder(s)                                  | Microsoft                                              |
| Final Report Date                                   | 10/04/2022                                             |
| Project Director/Principal Investigator Information |                                                        |
| Full Name                                           | Jayson Lusk                                            |
| Email                                               | <a href="mailto:jlusk@purdue.edu">jlusk@purdue.edu</a> |
| Phone                                               | 765-494-1903                                           |
| Grantee Organization Information                    |                                                        |
| Organization Name                                   | Purdue University                                      |
| Address                                             | 155 S Grant Street                                     |
| City, State Zip                                     | West Lafayette, IN, 47907-2114                         |
| Tax ID                                              | 35-6002041                                             |

| <b>Authorized Signing Official (ASO) Information</b> |                                |
|------------------------------------------------------|--------------------------------|
| Name                                                 | Jason Spall                    |
| ASO Title                                            | Assistant Director, Post-Award |
| Email                                                | spsacctmgmt@purdue.edu         |
| Phone                                                | 765-494-1055                   |

## General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

West Lafayette, Indiana, 4<sup>th</sup> Congressional District

- 1.2. How many new jobs were created because of this grant funding?

None

- 1.3. How many jobs were maintained because of this grant funding?

None

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No

## 2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

It is important to have real-time information on impacts of food supply chain disruptions so

that policy makers and industry leaders know where and how to respond and so that the public and media can better grasp the causes and nature of changes in food prices and availability being experienced at the retail level. At present, however, the relevant data is either non-existent, scattered across different federal agencies and private companies, or is in formats that is not easily accessible or digestible by decision makers and the interested public. The over-arching goal of the proposed project is to create a series of open-access online dashboards that help quantify and illustrate potential disruptions to the food supply chain, with a particular focus on COVID19 but with an eye toward flexibility toward any potential market disruptions. Ten different data dashboards have been created in four categories: prices, supply and production, consumer spending, and consumer preferences; all are available here: <https://ag.purdue.edu/cfdas/data-resources/dashboards/>. The intended outcomes of this product are to improve public understanding of the origins of their food and to improve policy and firm-level decision making so that food supply chain disruptions can be identified before reaching catastrophic levels.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

Initial dashboards focused on estimating the prevalence of COVID-19 among farmers and farmworkers and among food manufacturing workers in the United States by coupling county-level data on the number of farm or food processing workers relative to the general population with real-time data on confirmed COVID-19 cases and deaths. These dashboards identified “hot spots” and provided information potential supply disruptions and helped target resources. We also created a food industry vulnerability index to characterize the vulnerability of each of nine food industries in all 50 U.S. states are to disruptions from upstream industries (including labor). These results were communicated in peer-reviewed academic publications and in an online data dashboard and should help state planners and agribusinesses identify where they are most exposed in terms of input purchases. Additional data dashboards were created to better communicate retail food prices changes, an issue that has become particularly pronounced given the rise in inflation during 2022. These data have been useful in helping consumers understand which food items are becoming more or less expensive and have helped agribusinesses understanding the factors driving the increase in overall consumer food price index.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:
  - a) potential outcomes, i.e., findings, results, or recommendations that

- could impact U.S. agriculture and food systems, if used;
- b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

There are a number of potential outcomes. Analyses and dashboards have identified certain counties and industries that are potentially more vulnerable to COVID19 or supply chain disruptions than others. Local agribusinesses or policy makers could utilize the information to make investments to invest resources or diversify input purchases.

There are a variety of intermediate outcomes that resulted from presentations of the research and tools for various audiences that work on supply chain resiliency. For example, a presentation of the work was given before the U.S. Cyberstructure and Infrastructure Security Agency Food and Agriculture Sector Joint Membership Meeting in April 2022, and the PD had follow up meetings with members from several agencies (e.g., FDA and USDA) as well as food companies. Similar presentations have been given in a variety of settings, influencing additional work and outreach in this space including presentations to the Johns Hopkins Bloomberg School of Public Health and the National Coalition for Food and Agricultural Research (NCFAR), Lunch-N-Learn Hill Research Seminar Series, among others.

We are not aware of any end outcomes.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

The primary objectives of this project were to build out and improving a real-time preliminary dashboard that aims to communicate and illustrate to the broader public the nature of U.S. food supply. Additional specific objectives were to: 1) expand the current dashboard to add specific fruits and vegetables, 2) improve the current dashboard by adding metrics of potential losses in real-time dollar terms and by improving county-level estimated of agricultural workers impacted, 3) create a new dashboard focused on the geographic location of food processing and COVID cases, 4) create a new dashboard focused on real-time commodity specific production data to determine potential changes and reduction in supply from baseline production estimates, 5) create a new dashboard focused on retail prices and availability, and 6) create curated, open access datasets used in the dashboards to increase transparency and potential collaboration. Each of these

objectives have been accomplished.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

No.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Ten different data dashboards have been created in four categories: prices, supply and production, consumer spending, and consumer preferences; all are available here: <https://ag.purdue.edu/cfdas/data-resources/dashboards/>.

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

Much of the research was peer reviewed; the research resulted in two peer-reviewed journal articles. Many of the dashboards rely on "gold standard" government data, albeit combined and manipulated in new ways.

- 2.8. List key stakeholders that could be served by results of this project.

Policy makers; agribusinesses; media.

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of

information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write “nothing to report.” A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

As indicated, ten different data dashboards have been created in four categories: prices, supply and production, consumer spending, and consumer preferences; all are available here: <https://ag.purdue.edu/cfdas/data-resources/dashboards/> .

There have been several press releases associated with the work (e.g., <https://www.purdue.edu/newsroom/releases/2021/Q1/purdue-developed-dashboards-will-offer-timely-food-supply-info-during-crises.html>; <https://foundationfar.org/news/ffar-funds-dashboards-to-track-food-supply-chain-vulnerabilities/>)

Hundreds of media interviews have been conducted over the project period and many have mentioned work covered by this grant. Purdue Communications calculates that from 2020-2022, Lusk has garnered 4,973 media placements reaching an audience of 4.9 billion people. A few prominent examples of media mentions include:

<https://www.nytimes.com/2021/03/01/us/coronavirus-vaccine-farmworkers-california.html>  
<https://www.nytimes.com/2020/10/19/us/coronavirus-tomato-migrant-farm-workers.html>  
<https://www.washingtonpost.com/business/2020/09/24/seasonal-farm-workers-coronavirus/>  
<https://www.wsj.com/articles/for-farmers-and-consumers-a-crazy-year-in-food-11608071329>  
<https://www.latimes.com/science/newsletter/2021-03-10/indigenous-cielo-language-vaccine-access-coronavirus-today>  
<https://www.pbs.org/wgbh/frontline/article/undocumented-workers-farmworker-vaccine-priority-big-ag-states/>

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

There were no significant unexpected challenges.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state “Nothing to Report.” For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to



500-word limit)

Nothing to report.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

The project helped fund the time of a graduate student research assistant, who contributed to the data analysis.

### 3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.
- 3.2. Please provide a list of citations for the information products produced during the project.

Lusk and Chandra (2021) (see 3.3 for reference) has been cited 34 times by other academic researchers:

[https://scholar.google.com/scholar?cites=16182040336340761645&as\\_sdt=800005&sciodt=0,15&hl=en](https://scholar.google.com/scholar?cites=16182040336340761645&as_sdt=800005&sciodt=0,15&hl=en)

Wahdat and Lusk (forthcoming (see 3.3 for reference) has been cited 1 time by other academic researchers:

[https://scholar.google.com/scholar?cites=10843512875551745696&as\\_sdt=800005&sciodt=0,15&hl=en](https://scholar.google.com/scholar?cites=10843512875551745696&as_sdt=800005&sciodt=0,15&hl=en)

As indicated previously, there have been many media citations of the work.

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Lusk, J.L. and Chandra, R., 2021. Farmer and farm worker illnesses and deaths



from COVID-19 and impacts on agricultural output. Plos one, 16(4), p.e0250621.

Wahdat, A.Z. and Lusk, J., forthcoming. The Achilles Heel of the US Food Industries: Exposure to Labor and Upstream Industries in the Supply Chain. American Journal of Agricultural Economics. <https://doi.org/10.1111/ajae.12331>

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

Ten different data dashboards have been created in four categories: prices, supply and production, consumer spending, and consumer preferences; all are available here: <https://ag.purdue.edu/cfdas/data-resources/dashboards/>.

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

Nothing to report.

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Nothing to report.

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

Each dashboard describes the data sources used.

## 4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and



imageries). Please list the data generated for the award.

Most of the data to be used in the project are government produced, and will be collected and combined with annotations that indicate the source and provide variable definitions. All data manipulations will occur using the Microsoft Power BI platform, and newly created variables and outputs will likewise be freely available to download at the project website.

Thus far, the project makes use of data that are all publicly available, but makes novel calculations based on these data. Data for one dashboard are illustrated in a public Power BI dashboard at:

<https://ag.purdue.edu/agecon/Pages/FoodandAgVulnerabilityIndex.aspx>. Data used in one of the peer-reviewed article are hosted in the Harvard Dataverse repository: <https://doi.org/10.7910/DVN/OOOO3G>

For this article, the following data are used:

COVID-19 confirmed case are from Johns Hopkins: <https://coronavirus.jhu.edu/us-map>

County-level total population is for 2019 from the U.S. Census Bureau:

<https://www.census.gov/data/datasets/time-series/demo/popest/2010s-counties-total.html>

County-level data on number of farmers is from the 2017 Census of Agriculture (measured as the numbers of “producers”): <https://quickstats.nass.usda.gov/>

County-level data on number of hired farm laborers is from the 2017 Census of Agriculture (measured as an expense associated with “labor, hired – number of workers”): <https://quickstats.nass.usda.gov/>

County-level production for each commodity is from the 2017 Census of Agriculture:

<https://quickstats.nass.usda.gov/>

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

Each of the 10 dashboards uses different datasets and are databases that relate the datapoints to each other in terms of time, location, industry, product, etc.

- 4.3. Please provide copies of relevant metadata records to support FFAR’s mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

Data used in the peer-reviewed article are available in the Harvard Dataverse data repository:

<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/OOOO3G> .



FOUNDATION FOR  
FOOD & AGRICULTURE  
RESEARCH

## Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Jayson Lusk

PI Signature:  Date: 10/11/2022

Authorized Signing Official (ASO) Name: Jason Spall

ASO Signature: \_\_\_\_\_ Date: \_\_\_\_\_